EXPERIMENT 6:

```
import os import numpy
as np import
matplotlib.pyplot as plt
import imageio.v2 as
imageio
%matplotlib inline

face_dir="Downloads/SkinColor/SkinColor/FamilyPhoto"
mask_dir="Downloads/SkinColor/SkinColor/GroundFamilyPhoto"

images=[]
masks=[]
names=[]

image_files=os.listdir(face_dir)
mask_files=os.listdir(mask_dir)

for img_file in image_files:
    name=os.path.splitext(img_file)[0]

    for mask_file in mask_files:
        mask_name=os.path.splitext(mask_file)[0]

        if name==mask_name:
            img_path=os.path.join(face_dir,img_file)
mask_path=os.path.join(mask_dir,mask_file)

            image=imageio.imread(img_path)
mask=imageio.imread(mask_path)

            images.append(image)
masks.append(mask)
            names.append(name)   # FIX ADDED

            break

print("Number of Face Images
Loaded:",len(images))

for i in range(len(images)):
    image=images[i]
mask=masks[i]
name=names[i]
mask_rgb=mask[:,:,:3]

    skin_pixels=np.all(mask_rgb==[255,255,255],axis=-1)
```

```
    skin_image=np.zeros_like(image)
skin_image[skin_pixels]=image[skin_pixels]

    total_pixels=skin_pixels.size
skin_count=np.sum(skin_pixels)

    skin_percentage=(skin_count/total_pixels)*100

    print("Image:",name)     print("Skin
Pixels:",skin_count)     print("Skin
Percentage:%2f%%"%skin_percentage)
    print("-------------------------")

    plt.figure(figsize=(12,8))

    plt.subplot(2,2,1)
plt.imshow(image)
plt.title("Original Image")
plt.axis("off")

    plt.subplot(2,2,2)
plt.imshow(mask)
plt.title("Mask")

    plt.subplot(2,2,3)
plt.imshow(image)
plt.imshow(mask,alpha=0.4)
plt.title("Overlay")

    plt.subplot(2,2,4)
plt.imshow(skin_image)
plt.title("Extracted Skin")

    plt.suptitle(name)
plt.show()
     if
i==2:
break
```

```
Number of Face Images Loaded: 46
Image: 07-c140-12family-red-rr-398h
Skin Pixels: 7574
Skin Percentage:6.007106%
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```
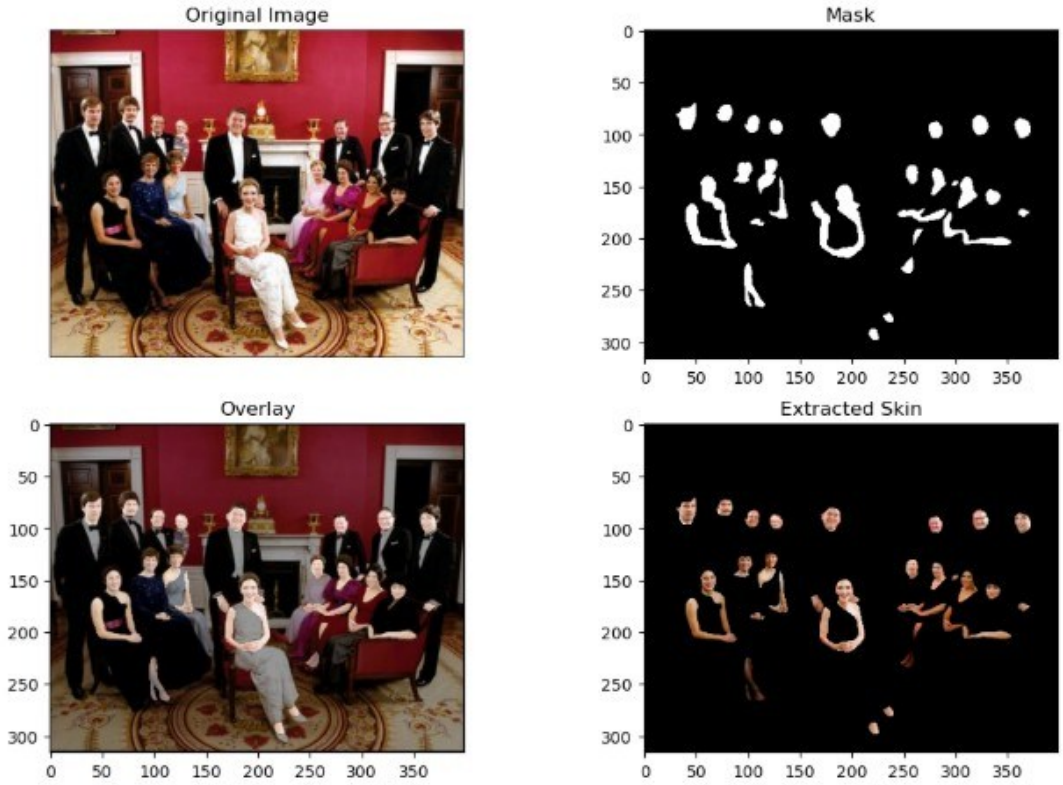
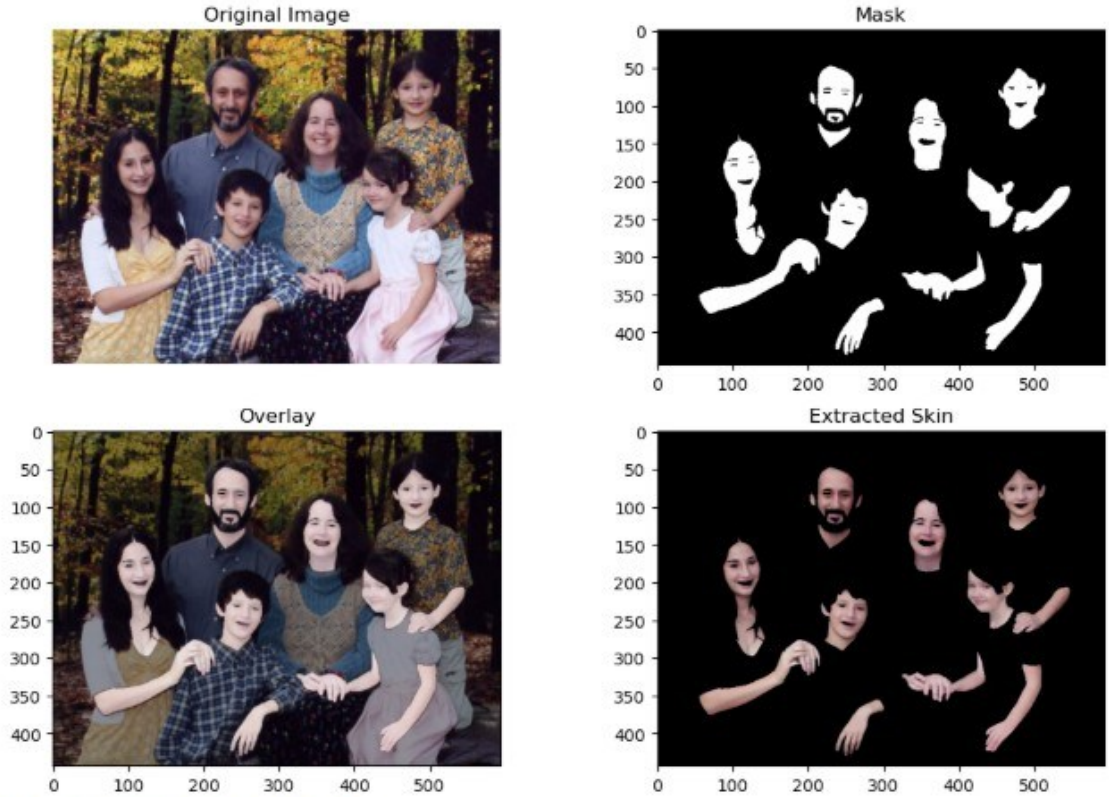




```
Image: 2007_family
Skin Pixels: 32066
Skin Percentage:12.185816%
```

2007_family




```
Image: 3115267-My-very-large-Indian-family-2
Skin Pixels: 24469
Skin Percentage:9.062593%
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```

3115267-My-very-large-Indian-family-2